

IN5340/IN9340

Detection theory

Lecture III

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Detection versus estimation

Detection:

The ability to discern between information-bearing patterns (signals) and random patterns that distract from the information (noise).

Used for (automated) scanning of (large) data sets for the purpose of detecting, localizing, and identifying particular targets of interest.

Estimation:

Extension of the detection part. It answers the question as to how much of a particular item of interest there is (i.e., a fine localization in the parameter space) and provides a measure of the accuracy or confidence.

Estimation

Some typical estimation examples are

- ▶ Frequency estimation (i.e. determine the actual Doppler shift).
- ▶ Differential time delay estimation (i.e. determine the distance or differential distance to a target).
- ▶ Amplitude estimation (i.e. target strength, size, distance).
- ▶ Phase and/or bandwidth estimation (i.e. target identification).
- ▶ Spectral estimation (i.e. target identification).
- ▶ Analog wave form estimation (i.e. original noise-free recovery of the original wave form using optimal filtering).

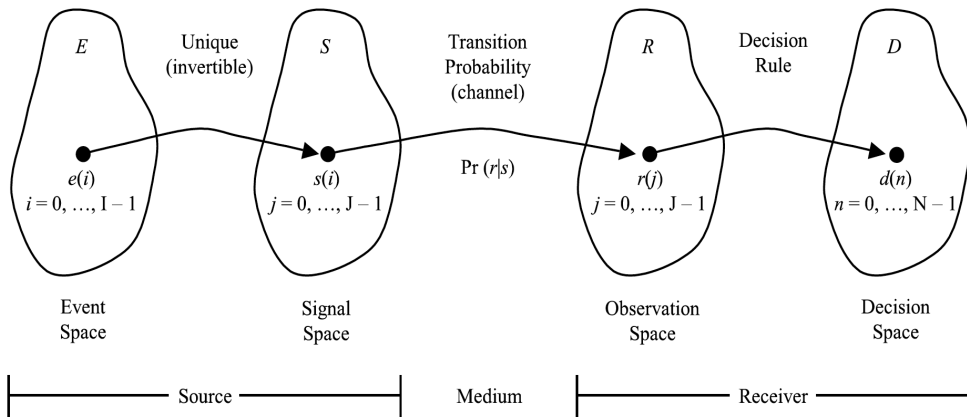
Detection

Some typical detection examples are

- ▶ Radar: passive, active, or bi-static (i.e. it uses an antenna and tests for target presence).
- ▶ Sonar: bi-static, active, or passive (i.e. it uses hydrophones and tests for target presence).
- ▶ Digital communication: coded binary words (symbols) (i.e. one wants to detect transmitted, possible encoded, message bits consisting of binary zeros and ones).
- ▶ Acoustic detection (i.e. it uses microphones): for intrusion alarm, emitter detection (i.e.gunnery, helicopter sound detection/localization etc.).
- ▶ Seismic detection (i.e. it uses geophones): to detect an earthquake, tunnel digging, nuclear testing etc.

Decision model

The spaces involved ...



Source: Detection theory : applications and digital signal processing / Ralph D. Hippenstiel; Figure 1.1.

Decision model (cont.)

E: The event space

Examples of events can be

- ▶ Messages: alphabetical, numeric, Morse code, etc.
- ▶ Parameters: frequency, phase angle, object moving or not, object speed, etc.
- ▶ Targets: target present/target absent, friend/foe, etc.

The message may be in a discrete or continuous form, may be a random variable, random vector, or a realization of a random process.

Example

$e_i, \quad i = 0, \dots, I - 1,$

- ▶ i.e. in the event space E, one of the I possible events can happen.
- ▶ We may or may not know the a priori probability $Pr\{e_i\}$ of the event e_i .

Decision model (cont.)

S: The signal space

The events are converted into representative signals.

Example (Event encoding, Ex 1.8)

Events:	Signals:
$A = e_0$	$\rightarrow s_0(t)$: signal representing event A
$B = e_1$	$\rightarrow s_1(t)$: signal representing event B
$C = e_2$	$\rightarrow s_2(t)$: signal representing event C
$\vdots$	$\vdots$
	for $0 \leq t \leq T$

Example (Intrusion alarm, Ex 1.12)

Events:	Signals:
e_0 (no intruder)	$\rightarrow s_0(t) = A \cos(\omega_0 t + \theta)$
e_1 (intruder present)	$\rightarrow s_1(t) = A \cos([\omega_0 + \omega_D(t)]t + \theta)$

where $\omega_d(t)$ is a time dependent frequency shift.

Decision model (cont.)

R: The observer space

The signals are observed (i.e. obtained) after having passed through some channel (i.e. the medium). In general, the received data is the transmitted signal plus an additive noise component:

$$r(t) = s(t) + n(t)$$

Example (Target detection, Ex 1.13)

Events:

Received data:

e_0 (no target) $\rightarrow r_0(t) = n(t)$, for $0 \leq t \leq T$.

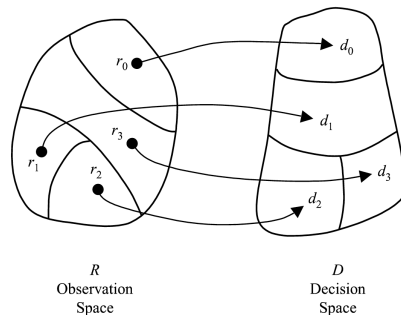
e_1 (target present) $\rightarrow r_1(t) = A s(t - t_R) + n(t)$, for $t_R \leq t \leq T + t_R$,

where A is the signal loss coefficient and t_R the propagation delay. (No Doppler shift assumed).

Decision model (cont.)

D: The decision space

- Usually: $\dim(D) = \dim(E)$
- We will focus on:
How to obtain a mapping from R to D .
- The mapping must be such that:
 - An unambiguous decision is made.
 - Every outcome must lead to some decision.



Example (Multiple decision, Ex 1.16)

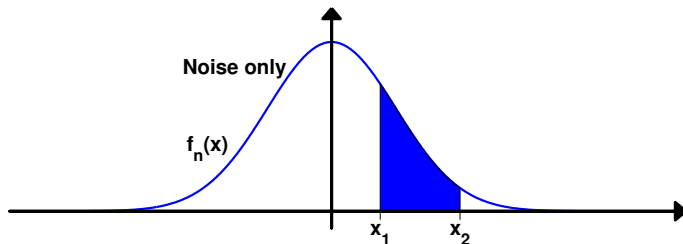
d_i : choose h_i ,
the hypothesis that the event e_i occurred, meaning that the observation lie in the space (region) R_i ,
for $i = 0, 1, 2, 3$

The Gaussian (normal) density function

$$\text{If } x \sim N(m_x, \sigma^2), \quad \text{then } f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-m_x)^2}{2\sigma^2}}$$

- ▶ Pdf symmetric about the mean m_x .
- ▶ Most likely value $x == \text{mean}$.
- ▶ Pdf completely described by mean and variance.
- ▶ x Independent $\Leftrightarrow x$ uncorrelated.
- ▶ x is closed under scalar multiplication and typical linear transformations.
This means that filtering, convolution, integration, differentiation, all common transforms (Hilbert, Fourier, Laplace, Z-transform) retain the characteristics of Gaussian random variables.
- ▶ Central limit theorem.
- ▶ This model fits many physical phenomena.

Probability



- ▶ The probability that a single random value would be between x_1 and x_2 (given the probability density function $f_X(x)$):

$$P(x_1 < x < x_2) = \int_{x_1}^{x_2} f_X(x) dx$$

- ▶ For any pdf: $\int_{-\infty}^{\infty} f_X(x) dx = 1$.

- ▶ And if $x \sim N(m_x, \sigma^2)$:

- ▶ $P(m_x - \sigma < x < m_x + \sigma) = 0.6827$, $P(m_x - 2\sigma < x < m_x + 2\sigma) = 0.9545$
- ▶ $P(m_x - 3\sigma < x < m_x + 3\sigma) = 0.9973$

Right tail probability

- ▶ Given the **standard normal distribution** (zero-mean, unit variance):

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}.$$

- ▶ Then the **cumulative distribution** is:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}}.$$

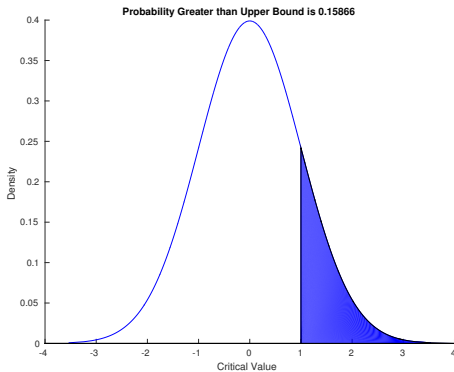
- ▶ The **probability of exceeding a given value**, $Q(x)$, is given by the **complementary cumulative distribution**:

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-\frac{t^2}{2}} = 1 - \Phi(x) = P_{FA}.$$

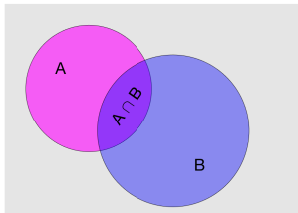
- ▶ Q is invertible (monotonically decreasing):
 $\gamma = Q^{-1}(P_{FA}).$

- ▶ In matlab:

```
p = normspec([-Inf,1],0,1,'outside')
```



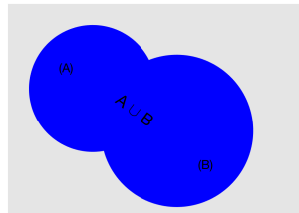
Set operations in probability theory



- ▶ Union: $A \cup B$
- ▶ Intersection: $A \cap B$
- ▶ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- ▶ Conditional probabilities:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$



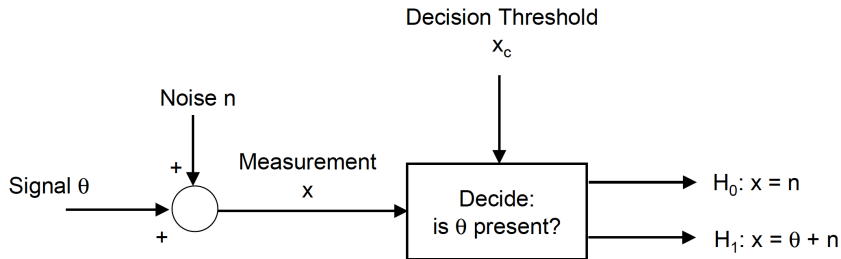
Bayes' rule:

▶ From $P(B | A) = \frac{P(A \cap B)}{P(A)} \Rightarrow$
 $P(A \cap B) = P(B | A) P(A)$

▶ Then:

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B | A) P(A)}{P(B)}$$

A basic signal detection problem

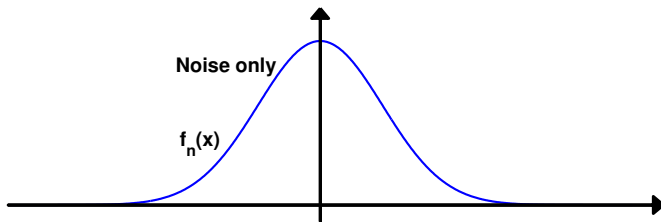


In a basic problem, we would have:

- ▶ A single known value of the signal θ .
- ▶ A random zero-mean normally distributed additive noise n .
- ▶ A single observable measurement x .
- ▶ To decide either H_0 or H_1 .

A basic signal detection problem

The noise model

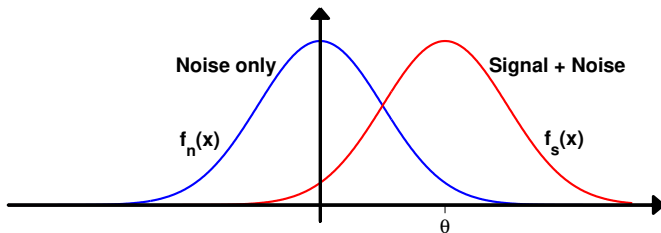


- ▶ $f_n(x)$: probability density function of the **noise**.
- ▶ The noise probability function can be arbitrary.
- ▶ A 'common' one is **normally-distributed, zero mean** (σ = standard deviation):

$$f_n(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$$

A basic signal detection problem

Signal + Noise



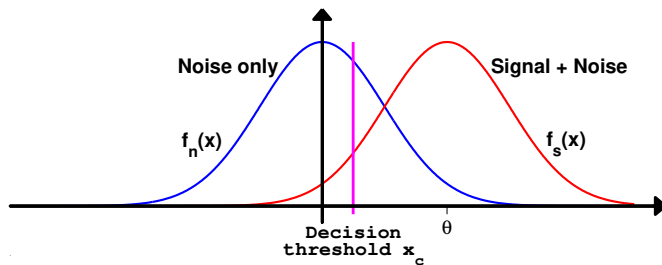
- ▶ $f_s(x)$: probability density function of **signal + noise**
- ▶ Could be arbitrary but often **normally-distributed**, mean θ (σ = standard deviation), i.e.

$$f_s(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta)^2}{2\sigma^2}}$$

- ▶ We define **signal-to-noise ratio (SNR)** as: $d' = \frac{\theta}{\sigma}$.
 - ▶ I.e. as the separation between **noise-only** and **signal+noise** measured in **standard deviations**.
 - ▶ if $\sigma = 1$, then $d' = \theta$.

A basic signal detection problem

Thresholded decision

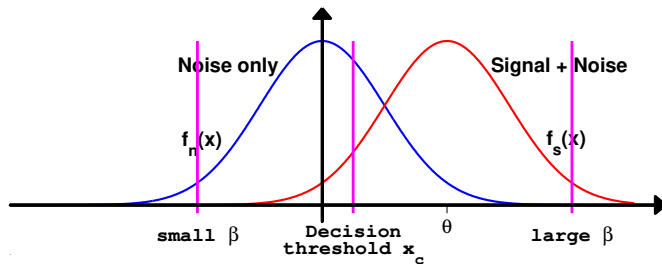


Decision criterion:

- ▶ if $x > x_c$, decide H_1 (signal + noise)
- ▶ Otherwise, decide H_0 (noise only)

A basic signal detection problem

Decision Bias, β



Decision Bias:

- We define **the decision Bias**, β , as

$$\beta = \frac{f_s(x_c)}{f_n(x_c)}.$$

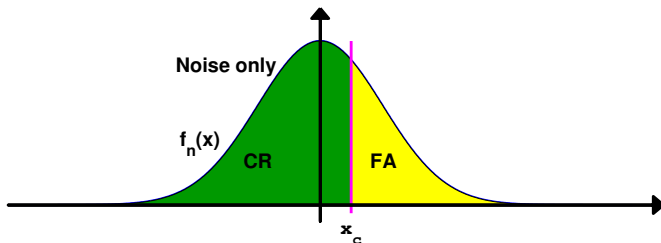
- I.e as the relative likelihood of observing the threshold value x_c if the signal is present to observing x_c if the signal is not present.

Decision outcomes

		Decision	
		Noise only (H_0)	Signal+Noise (H_1)
True state	Noise only H_0	Correct Rejection (CR) True negative (TN)	False Alarm (FA) Type I Error False Positive (FP)
	Signal+Noise H_1	Missed Detection (M) Type II Error False Negative (FN)	Correct Detection (D) True positive (TP)

Decision outcomes

Noise only



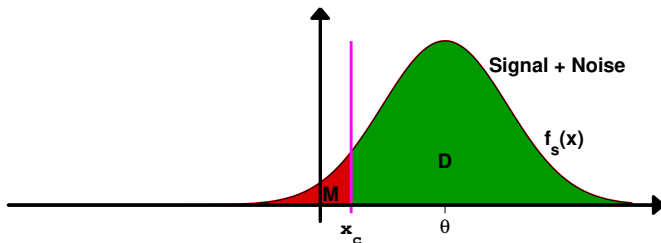
- If truly noise only, then

$$P(FA|N) = \int_{x_c}^{\infty} f_n(x) dx \quad \text{and} \quad P(CR|N) = \int_{-\infty}^{x_c} f_n(x) dx$$

- ... and we want x_c as large as possible.

Decision outcomes

Signal + Noise



- If signal is truly present, then

$$P(CD|S) = \int_{x_c}^{\infty} f_s(x) dx \quad \text{and} \quad P(MD|S) = \int_{-\infty}^{x_c} f_s(x) dx$$

- ... and we want x_c as small as possible.

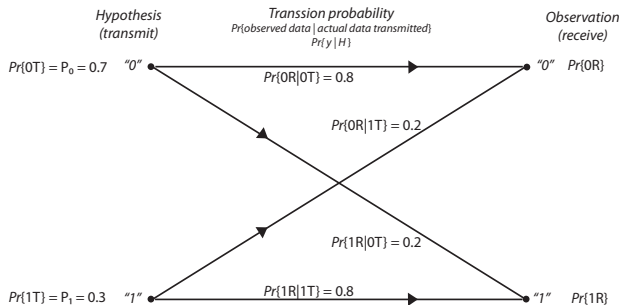
"a priori" and "a posteriori" probability

(... all these probabilities!)

- ▶ Suppose two possible events; binary "one" and "zero".
- ▶ The probability of event ' i ' (i.e. hypothesis H_i) given observation ' y ' is described by the **a posteriori probability** $Pr\{H_i|y\}$.
- ▶ The **a priori probability** of event ' i ' (i.e. the probability of hypothesis H_i) is given by $Pr\{H_i\}$.
- ▶ i.e. **a priori** is something we know (or can know) on the forehand, while **a posteriori** is something we have to experience.

"a priori" and "a posteriori" probability

(... all these probabilities!)



- ▶ The quantity of interest is the probability: $Pr\{ \text{"(actual) symbol transmitted"} | \text{"observed symbol"} \}$
- ▶ Find this by using Bayes' rule:
$$Pr\{H_i | y\} = \frac{Pr\{y | H_i\}}{Pr\{y\}} Pr\{H_i\}$$

"a priori" and "a posteriori" probability

(... all these probabilities!)

Example (Binary transmission)

- ▶ $Pr\{0R\} = Pr\{0R|1T\}Pr\{1T\} + Pr\{0R|0T\}Pr\{0T\}$
 $= 0.2 \times 0.3 + 0.8 \times 0.7 = 0.62$
- ▶ $Pr\{1R\} = Pr\{1R|0T\}Pr\{0T\} + Pr\{1R|1T\}Pr\{1T\}$
 $= 0.2 \times 0.7 + 0.8 \times 0.3 = 0.38$
- ▶ $Pr\{1T|1R\} = Pr\{1R|1T\}Pr\{1T\}/Pr\{1R\}$
 $= 0.8 \times 0.3 / 0.38 = 0.632$
- ▶ $Pr\{0T|0R\} = Pr\{0R|0T\}Pr\{0T\}/Pr\{0R\}$
 $= 0.8 \times 0.7 / 0.62 = 0.903$
- ▶ And
- ▶ $Pr\{1T, 1R\} = Pr\{1T|1R\}Pr\{1R\} = 0.632 \times 0.38 = 0.24$
- ▶ $Pr\{0T, 0R\} = Pr\{0T|0R\}Pr\{0R\} = 0.903 \times 0.62 = 0.56$

Bayes' detection

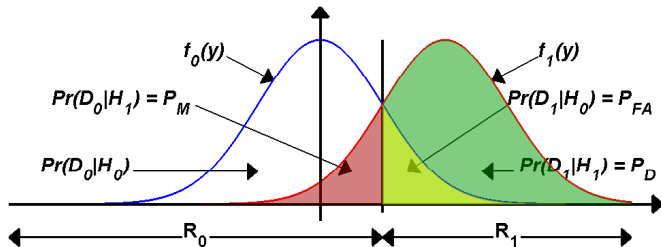
Requirements

Given:

- The observations: $y_n = s_n + n_n$ for $n = 1 \dots N$, i.e. $\mathbf{y} = (y_1, y_2, \dots, y_N)^T$

Requirements:

- The likelihood function $f_{\mathbf{y}|\mathbf{H}_i}(\mathbf{y}|\mathbf{H}_i) = f_i(\mathbf{y})$
= the density of $\mathbf{y}$ under the i 'th hypothesis, for $i = 0, 1$.
- The prior probability $Pr\{\mathbf{H}_i\}$ as P_i .
- Payoff matrix:
Cost associated with choosing hypothesis " i " when hypothesis " j " is true: C_{ij} .



Bayes' detection

Details

► Possible outcomes:

- (a) Choose H_0 ; H_0 is true (correct)
- (b) Choose H_1 ; H_0 is true (mistake)
- (c) Choose H_1 ; H_1 is true (correct)
- (d) Choose H_0 ; H_1 is true (mistake)

► The costs for the problem:

(C_{ij} : choosing "i" when "j" is true)

- (a) C_{00} (choose 0, 0 is true)
- (b) C_{10} (choose 1, 0 is true)
- (c) C_{11} (choose 1, 1 is true)
- (d) C_{01} (choose 0, 1 is true)

► The average cost (risk of penalty function):

$$\begin{aligned} C = & C_{00}Pr\{\text{choose 0, 0 is true}\} \\ & + C_{10}Pr\{\text{choose 1, 0 is true}\} \\ & + C_{11}Pr\{\text{choose 1, 1 is true}\} \\ & + C_{01}Pr\{\text{choose 1, 1 is true}\} \end{aligned}$$

► Using Bayes' rule, this becomes

$$\begin{aligned} C = & C_{00}P_0Pr\{\text{choose 0} \mid 0 \text{ is true}\} \\ & + C_{10}P_0Pr\{\text{choose 1} \mid 0 \text{ is true}\} \\ & + C_{11}P_1Pr\{\text{choose 1} \mid 1 \text{ is true}\} \\ & + C_{01}P_1Pr\{\text{choose 1} \mid 1 \text{ is true}\} \end{aligned}$$

► I.e.

$$\begin{aligned} C = & C_{00}P_0(1 - P_{FA}) + C_{10}P_0P_{FA} \\ & + C_{11}P_1P_D + C_{01}P_1P_M \end{aligned}$$

Bayes' detection

Details (continuing)

► The average cost:

$$C = C_{00}P_0(1 - P_{FA}) + C_{10}P_0P_{FA} \\ + C_{11}P_1P_D + C_{01}P_1P_M.$$

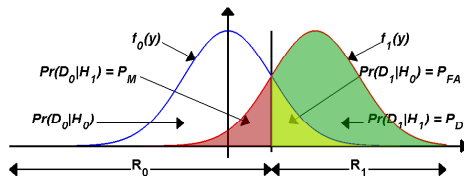
► can be expressed as:

$$C = C_{00}P_0 \int_{R_0} f_0(y) dy \\ + C_{10}P_0 \int_{R_1} f_0(y) dy \\ + C_{11}P_1 \int_{R_1} f_1(y) dy \\ + C_{01}P_1 \int_{R_0} f_1(y) dy.$$

► And by using

$R = R_1 \cup R_0$ where $R_1 \cap R_0 = \emptyset$, and

$$\int_R f_i(y) dy = 1 \text{ for } i = 0, 1$$



► it can be expressed as

$$C = C_{00}P_0 \int_{R_0} f_0(y) dy \\ + C_{10}P_0 \int_{R-R_0} f_0(y) dy \\ + C_{11}P_1 \int_{R-R_0} f_1(y) dy \\ + C_{01}P_1 \int_{R_0} f_1(y) dy.$$

Bayes' detection

Details (continuing ...)

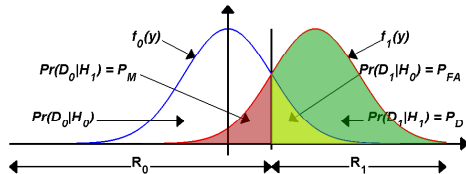
- The average cost reduces to:

$$\begin{aligned}
 C = & C_{00}P_0 \int_{R_0} f_0(y) dy \\
 & + C_{10}P_0 \left(1 - \int_{R_0} f_0(y) dy\right) \\
 & + C_{11}P_1 \left(1 - \int_{R_0} f_1(y) dy\right) \\
 & + C_{01}P_1 \int_{R_0} f_1(y) dy.
 \end{aligned}$$

- This can be rewritten as:

$$\begin{aligned}
 C = & \overbrace{C_{10}P_0 + C_{11}P_1}^{\text{fixed cost}} + \dots \\
 & \underbrace{\int_{R_0} [P_1(C_{01} - C_{11}) f_1(y) - P_0(C_{10} - C_{00}) f_0(y)] dy}_{\text{variable cost to be optimized}}.
 \end{aligned}$$

- I.e. optimal value depends on R_0



- Assuming cost of a bad decision is always larger than the cost of a good one, i.e.

$$C_{10} - C_{00} > 0 \text{ and } C_{01} - C_{11} > 0$$

- then all terms in the variable cost are positive.
- Optimal choice of R_0 : Assign y to R_0 when negative term in integral is dominant!
- Conversely: Assign y to R_1 when first term in integral is dominant; hence no additional cost is added to the fixed cost.

Bayes' detection

- ▶ The decision criterion becomes:
$$P_1 [C_{01} - C_{11}] f_1(\mathbf{y}) \underset{H_0}{\overset{H_1}{>}} P_0 [C_{10} - C_{00}] f_0(\mathbf{y})$$
- ▶ **The Bayes' formulation:**
$$\Lambda(\mathbf{y}) = \frac{f_1(\mathbf{y})}{f_0(\mathbf{y})} \underset{H_0}{\overset{H_1}{>}} \frac{P_0 [C_{10} - C_{00}]}{P_1 [C_{01} - C_{11}]} = \lambda$$
- ▶ $\Lambda(\mathbf{y}) = \frac{f_1(\mathbf{y})}{f_0(\mathbf{y})}$ is known as the likelihood ratio (LR).
- ▶ Interpretation:
 - ▶ If $\Lambda(\mathbf{y}) > \lambda$, decide on H_1 .
 - ▶ If $\Lambda(\mathbf{y}) < \lambda$, decide on H_0 .
- ▶ Note: Both sides are non-negative, i.e. the inequality is preserved when taking the logarithm of both sides (or another monotonic function)
 - ▶ $\ln \Lambda(\mathbf{y}) \underset{H_0}{\overset{H_1}{>}} \ln \lambda$
 - ▶ $\ln \Lambda(\mathbf{y})$ is known as the log likelihood ratio (LLR).

Maximum a posteriori (MAP) detection

- ▶ We have prior probabilities but either cannot, or don't want to, assign a cost for making the possible decisions, i.e. the maximum a posteriori (MAP) detection philosophy.
- ▶ Start with the dominant **a posteriori** probability:

$$Pr(H_1 | y) \underset{H_0}{\overset{H_1}{>}} Pr(H_0 | y)$$

- ▶ i.e. given y , which of H_0 and H_1 is more likely true?
- ▶ Rewrite in incremental form

$$Pr(H_1 | y < y \leq y + dy) \underset{H_0}{\overset{H_1}{>}} Pr(H_0 | y < y \leq y + dy)$$

- ▶ Bayes' theorem gives

$$Pr(H_i | y < y \leq y + dy) = \frac{Pr(y < y \leq y + dy | H_i)}{Pr(y < y \leq y + dy)} Pr(H_i), \quad i = 1, 2.$$

- ▶ Using (when dy becomes small)

$$Pr(y < y \leq y + dy | H_i) = f_i(y) dy, \text{ and} \\ Pr(y < y \leq y + dy) = f(y) dy.$$

- ▶ we get

$$Pr(H_i | y) = \frac{f_i(y)}{f(y)} Pr(H_i), \quad i = 1, 2.$$

- ▶ and the MAP detector

$$\frac{f_1(y)}{f_0(y)} \underset{H_0}{\overset{H_1}{>}} \frac{Pr(H_0)}{Pr(H_1)} = \frac{P_0}{P_1}.$$

Maximum likelihood (ML) criterion

- ▶ If no prior probability nor any cost information is available, then comparing the likelihood function will give a meaningful detection scheme.
- ▶ That is: "Given the data, which conditional PDF (i.e. $f_0(\mathbf{y})$ or $f_1(\mathbf{y})$) has more likely generated the data?"
- ▶ In the scalar case, the detection scheme becomes

$$f_1(y) \underset{H_0}{\overset{H_1}{>}} f_0(y)$$

- ▶ and for multiple observations

$$\frac{f_1(\mathbf{y})}{f_0(\mathbf{y})} \underset{H_0}{\overset{H_1}{>}} 1$$

Minimum probability of error criterion

- ▶ The criterion is used in binary communication problems where the cost of making an error is the same and the cost of making a correct decision is zero.
- ▶ We do allow non-equal prior probabilities (depends on the source coding).
- ▶ We set up the average cost in the same way as done in the Bayes' case.
- ▶ The average cost becomes ($C_{00} = C_{11} = 0$, $C_{10} = C_{01} = 1$)

$$C = P_0 \int_{R_1} f_0(\mathbf{y}) d\mathbf{y} + P_1 \int_{R_0} f_1(\mathbf{y}) d\mathbf{y}$$

- ▶ The Bayes' approach then gives

$$\Lambda(\mathbf{y}) = \frac{f_1(\mathbf{y})}{f_0(\mathbf{y})} \underset{H_0}{\overset{H_1}{>}} \frac{P_0}{P_1}$$

- ▶ This is the same expression (including the threshold) as that of the MAP detector!

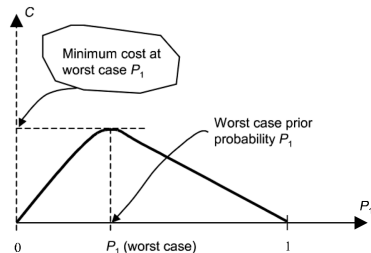
Min-Max criterion

- ▶ The cost information (i.e. C_{ij} , $i, j = 0, 1$) is available but not the **a priori** probabilities P_0 or P_1 .
- ▶ The best Bayes' detection scheme is then desirable: The worst possible choice of H_i , $i = 0, 1$ (Murphy's law)
- ▶ The average Bayes' cost (using P_{FA} and P_M):

$$C = P_0 C_{00} + P_1 C_{11} + \dots$$

$$P_1(C_{00} - C_{11})P_M + P_0(C_{10} - C_{00})P_{FA}$$

- ▶ When $P_0 = 1$, then $P_1 = P_{FA} = 0$ and $C = C_{00}$
- ▶ When $P_1 = 1$, then $P_0 = P_{FA} = P_M = 0$ and $C = C_{11}$



- ▶ If we assume $C_{00} = C_{11} = 0$ (as in most real life situations), we can plot C as a function of P_1 and either get a straight line or a curve as in the figure.
- ▶ In the figure, the maximum occurs when the slope of the curve is zero.
- ▶ To find the maximum, set the partial derivative of C with respect to P_1 to zero.

Min-Max criterion ...

- So given the maximum average cost, we minimize it, hence the name Min-Max criterion

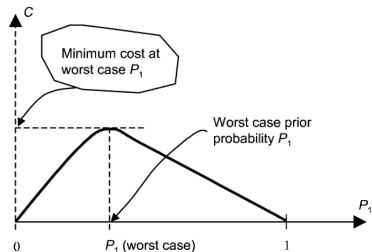
$$\frac{\partial C}{\partial P_1} = \frac{\partial C}{\partial P_1} \left\{ (1 - P_1)C_{00} \cdots \right. \\ \left. + P_1 C_{11} + P_1 (C_{00} - C_{11})P_M \cdots \right. \\ \left. + (1 - P_1)(C_{10} - C_{00})P_{FA} \right\}$$

- Setting it to zero gives:

$$(C_{11} - C_{00}) + (C_{01} - C_{11})P_M \\ - (C_{10} - C_{00})P_{FA} = 0$$

- If $C_{00} = C_{11} = 0$ we get

$$C_{01}P_M = C_{10}P_{FA}$$



- The detector form using the Bayes' cost formula is the Bayes detector with a threshold λ that requires the above equations to be true.

The Neyman-Pearson criterion

- ▶ One of the first in a class called CFAR (constant false alarm rate) detectors.
- ▶ Typically used in radar and sonar applications
 - ▶ Allows constant false alarm rate.
 - ▶ Attractive when loss of life or essential equipment are involved.
 - ▶ Desirable when resource allocations are important and the maximum false alarm rate has to be kept at a tolerable level.
- ▶ In essence, the false alarm rate is fixed while the detection probability is maximized.

▶ Approach:

- ▶ Fix the false alarm rate P_{FA} at a value α (i.e. force $(P_{FA} - \alpha)$ to zero) while maximizing P_D (or minimizing P_M since $P_M = 1 - P_D$).
- ▶ Easily solved with Lagrange multipliers.

▶ Cost function:

$$\begin{aligned} J &= \overbrace{\int_{R_0} f_1(\mathbf{y}) d\mathbf{y}}^{\text{minimize}} + \lambda \left[\overbrace{\int_{R_1} [f_0(\mathbf{y}) d\mathbf{y} - \alpha]}^{\text{equality constraint}} \right] \\ &= \int_{R_0} f_1(\mathbf{y}) d\mathbf{y} + \lambda \left[1 - \int_{R_0} [f_0(\mathbf{y}) d\mathbf{y} - \alpha] \right] \\ &= \lambda(1 - \alpha) + \int_{R_0} [f_1(\mathbf{y}) - \lambda f_0(\mathbf{y})] d\mathbf{y} \end{aligned}$$

The Neyman-Pearson criterion ...

- Solving the cost function:

$$J = \lambda(1 - \alpha) + \int_{R_0} [f_1(y) - \lambda f_0(y)] dy$$

- For fixed λ , the first term is constant (>0).
- Need only to consider

$$\int_{R_0} [f_1(y) - \lambda f_0(y)] dy$$

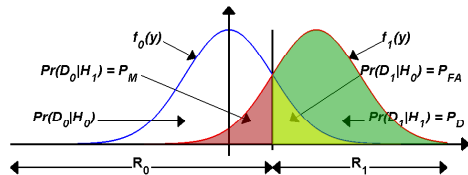
- Same as for the Bayes' detector.

- We get

$$f_1(y) \underset{H_0}{\overset{H_1}{>}} \lambda f_0(y)$$

- or

$$\Lambda(y) = \frac{f_1(y)}{f_0(y)} \underset{H_0}{\overset{H_1}{>}} \lambda$$



- By solving the above equation for our variable (here z), we get:

$$z \underset{H_0}{\overset{H_1}{>}} \gamma, \text{ and}$$

- The false alarm rate is then given by

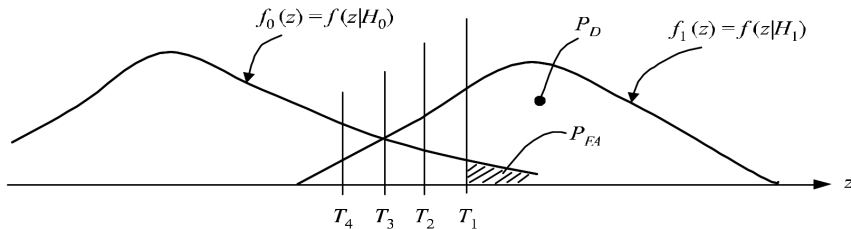
$$P_{FA} = \int_{\gamma}^{\infty} f_0(z) dz = \alpha$$

- The threshold γ is chosen so that the P_{FA} constraint is met (i.e. $P_{FA} = \alpha$).

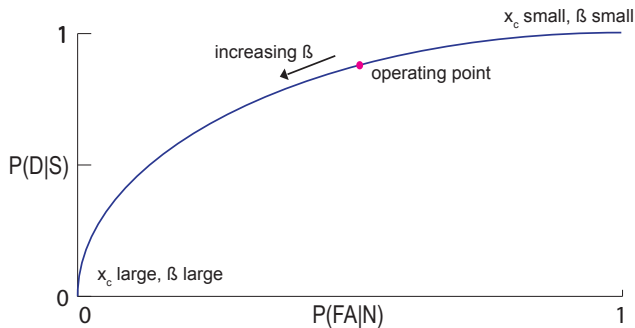
Receiver operator characteristic (ROC)

Detector performance evaluation

- ▶ Usually, we want to know
 - ▶ The probability of detection, P_D .
 - ▶ (The probability of miss, P_M , but $P_D + P_M = 1$).
 - ▶ The probability of false alarm, P_{FA} .
- ▶ By plotting P_D against P_{FA} we can compare the performance of different detectors.
- ▶ For each unique threshold, a unique pair of P_D and P_{FA} exists!

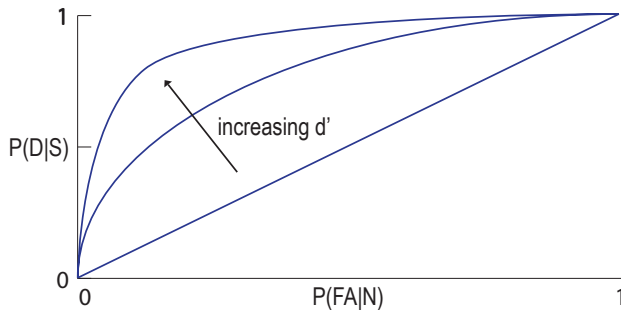


Receiver operator characteristic (ROC) (continuing)



- Generally: $\beta = \lambda = \frac{\partial P_{FA}}{\partial P_D}$, i.e. the slope is the threshold of the detector.
- Also, from Bayes' detector: $\beta = \frac{f_s(x_c)}{f_n(x_c)}$

Receiver operator characteristic (ROC) (continuing ...)



- **Signal-to-noise ratio (SNR):** $d' = \frac{\theta}{\sigma}$.
- A given value of d' traces out a unique ROC curve as β varies.

Terminology

Decision

		Decision	
		Noise only (H_0)	Signal+Noise (H_1)
True state	Noise only H_0	Correct Rejection (CR) True negative (TN)	False Alarm (FA) Type I Error False Positive (FP)
	Signal+Noise H_1	Missed Detection (M) Type II Error False Negative (FN)	Correct Detection (D) True positive (TP)

Terminology ...

From wikipedia.org

► **Sensitivity** or true positive rate (TPR):

$$TPR = TP/P = TP/(TP + FN)$$

► **Specificity** (SPC) or true negative rate:

$$SPC = TN/N = TN/(FP + TN)$$

► **Prediction** or positive predictive value (PPV):

$$PPV = TP/(TP + FP)$$

► Negative predictive value (NPV):

$$NPV = TN/(TN + FN)$$

► Fall-out or false positive rate (FPR):

$$FPR = FP/N = FP/(FP + TN) = 1 - SPC$$

► False negative rate:

$$FNR = FN/(TP + FN) = 1 - TPR$$

► False discovery rate (FDR):

$$FDR = FP/(TP + FP) = 1 - PPV$$

► Accuracy (ACC):

$$ACC = (TP + TN)/(P + N)$$

Terminology (From wikipedia.org)

		Condition (as determined by "Gold standard")			
		Total population	Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$
Test outcome	Test outcome positive	True positive	False positive (Type I error)	Positive predictive value (PPV, Precision) = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Test outcome positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Test outcome positive}}$
	Test outcome negative	False negative (Type II error)	True negative	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Test outcome negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Test outcome negative}}$
Positive likelihood ratio (LR+) = TPR/FPR		True positive rate (TPR, Sensitivity, Recall) = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR, Fall-out) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Negative likelihood ratio (LR-) = FNR/TNR		False negative rate (FNR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	True negative rate (TNR, Specificity, SPC) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$		
Diagnostic odds ratio (DOR) = LR+/LR-					

OBS: Flipped compared to previous foil

Sensitivity

- ▶ Consider the example of a medical test used to identify a disease.
- ▶ Sensitivity of the test is the proportion of people known to have the disease, who test positive for it:

$$\begin{aligned}\text{Sensitivity} &= \frac{\text{no. of true positives}}{\text{no. of true positives} + \text{no of false negatives}} \\ &= \frac{\text{no. of true positives}}{\text{total no. of sick individuals in population}} \\ &= \text{prob. of a positive test, given that the patient is ill}\end{aligned}$$

- ▶ A negative result in a test with high sensitivity is useful for ruling out disease.
- ▶ https://en.wikipedia.org/wiki/Sensitivity_and_specificity

Specificity

- ▶ Consider the example of a medical test for diagnosing a disease.
- ▶ Specificity of a test is the proportion of healthy patients known not to have the disease, who will test negative for it:

$$\begin{aligned}\text{Specificity} &= \frac{\text{no. of true negatives}}{\text{no. of true negatives} + \text{no of false positives}} \\ &= \frac{\text{no. of true negatives}}{\text{total no. of well individuals in population}} \\ &= \text{prob. of a negative test, given that the patient is well}\end{aligned}$$

- ▶ Positive result in a test with high specificity is useful for ruling in disease.
- ▶ https://en.wikipedia.org/wiki/Sensitivity_and_specificity

Sensitivity and Specifisity

From faktisk.no

Prevalens

Andel reelt smittede blant de som testes

10 %

Sensitivitet

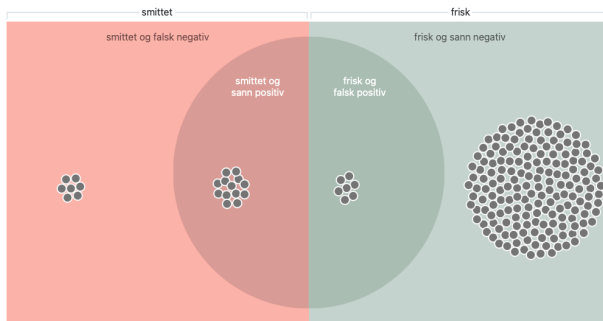
Hvor godt testen gjenkjenner smittede

63 %

Spesifisitet

Hvor godt testen gjenkjenner friske

96 %



Grafikken viser et nedskalert eksempel med 200 tester.

Utvikling: [Jarl Bakken](#)

Faktisk.

<https://www.faktisk.no/artikler/r8q/er-14-av-15-positive-koronaprover-falske>

Worked example

- ▶ Norwegian Directorate of Health:
 - ▶ *Evaluation of Abbot's Panbio COVID-19 rapid antigen test in Norway*, 3 December 2020.
 - ▶ <https://www.helsedirektoratet.no/rapporter/evaluation-of-abbots-panbio-covid-19-rapid-antigen-test-in-norway/>
 - ▶ From a total of 3991 cases, 250 where positive.

- ▶ The rapid covid test found
 - ▶ 186 out of the 250 positive.
 - ▶ 3738 out of the 3741 negative.
- ▶ This gives the following outcomes

True state	Decision	
	CR = TN = 3738	FA = FP = 3
	M = FN = 64	D = TP = 186

- ▶ This gives
 - ▶ Sensitivity, $TP / (TP + FN)$: 74.4 %
 - ▶ Specificity, $TN / (FP + TN)$: 99.9 %
- ▶ If we test 100 000 healthy, 100 of these would be found positive.
- ▶ If we test 1000 with prevalence of 4 ‰, we will miss out on one positive and fail on one negative.